

A Unified Control Plane for Machine-Learning Security Services

An Authenticated Gateway for Independently Released ML Security Products

github.com/poojakira/unified-ml-security-platform - public, default branch main.

76
Core Python tests passed

95.78%
Core statement coverage

4
Production synchronous service slots

Contract stubs
Local product topology

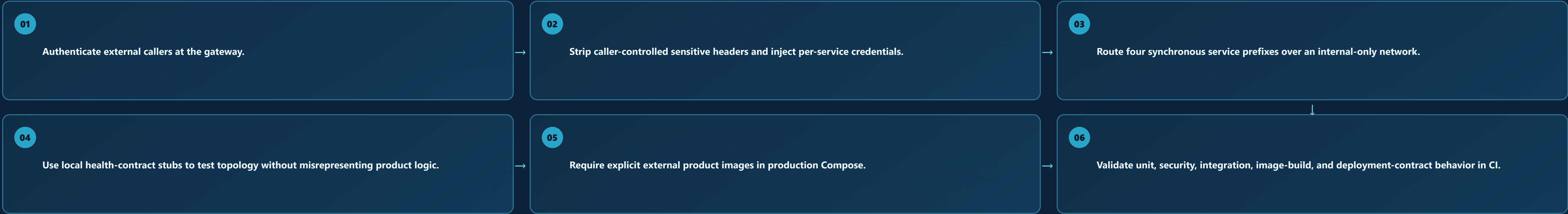
Contribution

A real authenticated FastAPI control plane with four production HTTP service slots, a local contract-test topology built from stubs, and a production Compose contract that requires externally released product images instead of pretending the stubs are product implementations.

Verified Evidence

- 76 tests passed.
- 95.78% statement coverage.
- Separate local contract-stub health jobs for MCP, LLM red-team, dataset-poison, and model-privacy completed successfully.
- Integration topology tests completed successfully.
- Security scan, dependency audit, production Compose validation, and gateway image build/push job completed successfully.
- Production product images: Externally supplied; stubs not published

Architecture / Threat Flow



Flow labels come directly from the repository research brief. They describe the tested control path, not a universal security guarantee.

Methodology and Control Path

Authenticate external callers at the gateway.	Strip caller-controlled sensitive headers and inject per-service credentials.
Route four synchronous service prefixes over an internal-only network.	Use local health-contract stubs to test topology without misrepresenting product logic.
Require explicit external product images in production Compose.	Validate unit, security, integration, image-build, and deployment-contract behavior in CI.

Limitations and Residual Risk

- Product business logic is owned by external repositories/images.
- No production deployment, uptime, or load SLO is established.
- Multi-replica shared rate limiting/state distribution requires deployment-specific infrastructure.
- Batch tools such as HF provenance scanning and adversarial robustness are intentionally not forced behind synchronous HTTP routes.

Reproducibility

Expected core Python evidence: 76 passed, 95.78% coverage.